

# Series with terms of variable sign

# Outline

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Alternating series and the Leibniz test

# Series with terms of variable sign



# Introduction: series with terms of variable sign

- ▶ So far we have studied series with nonnegative terms, for which we have the comparison, limit comparison, root and ratio tests
- ▶ We now consider series whose terms may change sign: these tests do not apply directly
- ▶ We introduce **absolute convergence**, which reduces the study of  $\sum a_k$  to that of the series with nonnegative terms  $\sum |a_k|$
- ▶ We will prove that absolute convergence implies convergence, but the converse does not hold

# Absolute convergence

## Definition 1: absolutely convergent series

*A series  $\sum a_k$  is absolutely convergent if the series  $\sum |a_k|$  converges.*

## Theorem 1

*If a series  $\sum a_k$  converges absolutely, then it converges.*

## Proof (1/2)

Without loss of generality, let  $n_0 = 0$  and consider the following series:

$$\sum_{k=0}^{\infty} (|a_k| - a_k)$$

It is a series with nonnegative terms since, for every  $k \in \mathbb{N}$ , we have:

$$\begin{cases} |a_k| - a_k = -2a_k \geq 0 & \text{if } a_k < 0 \\ |a_k| - a_k = 0 & \text{if } a_k \geq 0 \end{cases}$$

Moreover, by the triangle inequality, for every  $k \in \mathbb{N}$ , we have:

$$\underbrace{|a_k| - a_k}_{\geq 0} = ||a_k| - a_k| \leq |a_k| + |a_k| = 2|a_k|$$

## Proof (2/2)

Hence, by the comparison test for series with nonnegative terms, we have

$$\sum_{k=0}^{\infty} |a_k| \text{ convergent} \implies \sum_{k=0}^{\infty} (|a_k| - a_k) \text{ convergent}$$

Since for every  $k \in \mathbb{N}$  we have  $a_k = |a_k| - (|a_k| - a_k)$  and the series  $\sum_{k=0}^{\infty} |a_k|$  and  $\sum_{k=0}^{\infty} (|a_k| - a_k)$  converge, we have:

$$\sum_{k=0}^{\infty} a_k = \sum_{k=0}^{\infty} |a_k| - \sum_{k=0}^{\infty} (|a_k| - a_k)$$

that is, the series is the difference of two convergent series. Consequently, the series  $\sum_{k=0}^{\infty} a_k$  is convergent as well. □

## An alternative proof

We now see an alternative proof of the same theorem, based on splitting the partial sums into positive and negative terms.

## Alternative proof (1/2)

Without loss of generality, let  $n_0 = 0$  and split the sequence  $\{s_n\}$  of partial sums into two sequences, the first containing only the positive terms and the second only the negative terms:

$$s_n^+ = \sum_{\substack{k \in \{0,1,\dots,n\}: \\ a_k > 0}} a_k \quad \text{and} \quad s_n^- = \sum_{\substack{k \in \{0,1,\dots,n\}: \\ a_k < 0}} -a_k$$

$$\text{hence} \quad s_n = s_n^+ - s_n^-$$

Consequently, it is enough to show that the sequences  $\{s_n^+\}$  and  $\{s_n^-\}$  are convergent to conclude that  $\{s_n\}$  converges, and hence that the series  $\sum a_k$  converges.

## Alternative proof (2/2)

We observe that  $\{s_n^+\}$  and  $\{s_n^-\}$  are monotone non-decreasing sequences and moreover, by the triangle inequality, we have:

$$s_n^+ \leq \sum_{k=0}^n |a_k| \quad \text{and} \quad s_n^- \leq \sum_{k=0}^n |a_k|$$

On the other hand, by hypothesis the series  $\sum a_k$  converges absolutely, i.e.,  $\sum |a_k|$  converges, and hence the quantity  $\sum_{k=0}^n |a_k|$  is bounded; therefore the sequences  $\{s_n^+\}$  and  $\{s_n^-\}$  are bounded above and non-decreasing, and hence they converge, by the monotone sequence theorem. □

## Example: absolute convergence

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k^{\alpha}} \quad \text{with } \alpha > 1$$

We have:

$$\left| \frac{(-1)^k}{k^{\alpha}} \right| = \frac{1}{k^{\alpha}}, \quad \forall k \in \mathbb{N}, k > 1$$

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} \quad \text{is convergent for } \alpha > 1$$

Therefore the series converges absolutely and hence it converges.

# Absolute convergence and simple convergence

Absolute convergence implies ordinary convergence, also called simple convergence:

$$\sum |a_k| \text{ convergent} \quad \Rightarrow \quad \sum a_k \text{ convergent} \quad (1)$$

but the converse is not true (we will give a counterexample later):

$$\sum a_k \text{ convergent} \quad \nRightarrow \quad \sum |a_k| \text{ convergent} \quad (2)$$

Hence absolute convergence is a sufficient but not necessary condition for ordinary convergence.

# Summary

## Absolute convergence:

- ▶  $\sum a_k$  is absolutely convergent if the series with nonnegative terms  $\sum |a_k|$  converges
- ▶ To study it we use the tests for series with nonnegative terms

## Theorem:

- ▶ Absolute convergence  $\Rightarrow$  convergence (two proofs: comparison with  $\sum (|a_k| - a_k)$ , or partial sums  $s_n^+$  and  $s_n^-$ )
- ▶ The converse is false: absolute convergence is sufficient but not necessary

# Alternating series and the Leibniz test



# Introduction: alternating series and the Leibniz test

- ▶ Among series with terms of variable sign, a particularly simple case is given by alternating series, for which the following convergence test holds
- ▶ The **Leibniz test** allows us to establish convergence even when the series does **not** converge absolutely
- ▶ We will also see an estimate of the error made by approximating the sum with a partial sum, and several examples of application

## Theorem 2: Leibniz (alternating series) test

*Consider the series*

$$\sum_{k=n_0}^{\infty} (-1)^k a_k \quad \text{with} \quad a_k \geq 0, \forall k$$

*If the sequence  $\{a_k\}$  is decreasing and  $a_k \rightarrow 0$  as  $k \rightarrow +\infty$ , then the series is convergent. Moreover:*

$$s_{2n} = \sum_{k=n_0}^{2n} (-1)^k a_k \downarrow s \quad \text{and} \quad s_{2n+1} = \sum_{k=n_0}^{2n+1} (-1)^k a_k \uparrow s$$

as  $n \rightarrow +\infty$

- ▶ The partial sums with even index approximate the sum  $s$  from above and those with odd index from below.
- ▶ The Leibniz test can clearly be applied also if the terms eventually have alternating signs and the sequence  $\{a_k\}$  is eventually decreasing.

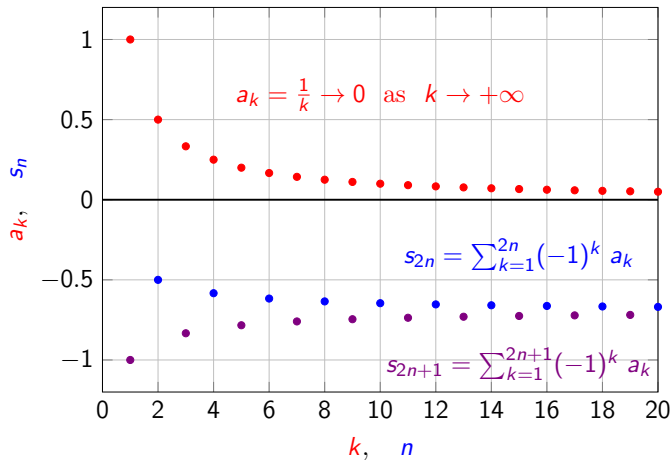
## Example: Leibniz test (1/2)

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} (-1)^k \frac{1}{k}$$

The sequence  $a_k = \frac{1}{k}$  is decreasing and non-negative. Moreover  $\frac{1}{k} \rightarrow 0$  as  $k \rightarrow +\infty$ , hence it satisfies the two conditions of the Leibniz test theorem. Consequently, the series is convergent.

## Example: Leibniz test (2/2)



## A counterexample

The series  $\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$  converges and is a counterexample to (2), that is, a series that converges but does not converge absolutely, since:

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^k}{k} \right| = \sum_{k=1}^{\infty} \frac{1}{k} \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges (harmonic series)}$$

## Proof of the Leibniz test (1/3)

Consider the sequence of partial sums  $\{s_n\}$  with  $n_0 = 0$  and the two subsequences extracted from it,  $\{s_{2n}\}$  and  $\{s_{2n+1}\}$ .

For  $\{s_{2n}\}$  we have:

$$s_0 = a_0, \quad s_2 = s_0 - a_1 + \underbrace{a_2}_{\leq a_1} \leq s_0, \quad s_4 = s_2 - a_3 + \underbrace{a_4}_{\leq a_3} \leq s_2, \quad \dots$$

hence the sequence  $\{s_{2n}\}$  is monotone decreasing.

For  $\{s_{2n+1}\}$  we have:

$$s_1 = a_0 - a_1, \quad s_3 = s_1 + a_2 - \underbrace{a_3}_{\leq a_2} \geq s_1, \quad s_5 = s_3 + a_4 - \underbrace{a_5}_{\leq a_4} \geq s_3, \quad \dots$$

hence the sequence  $\{s_{2n+1}\}$  is monotone increasing.

## Proof of the Leibniz test (2/3)

Moreover, we have:

$$s_1 \leq s_{2n+1} = s_{2n} - a_{2n+1} \leq s_{2n} \leq s_0$$

therefore  $\{s_{2n+1}\}$  is bounded above and  $\{s_{2n}\}$  is bounded below. The two sequences are hence convergent, by the monotone sequence theorem.

The two sequences converge to the same limit, because

$$0 \leq s_{2n} - s_{2n+1} \leq a_{2n+1} \rightarrow 0 \quad \text{as} \quad n \rightarrow +\infty$$

## Proof of the Leibniz test (3/3)

Calling  $s$  this limit, since  $\{s_{2n}\}$  is monotone decreasing and  $\{s_{2n+1}\}$  is monotone increasing, we have:

$$s_{2n} = \sum_{k=n_0}^{2n} (-1)^k a_k \downarrow s \quad \text{and} \quad s_{2n+1} = \sum_{k=n_0}^{2n+1} (-1)^k a_k \uparrow s$$

$$\text{as } n \rightarrow +\infty$$

Hence the series is convergent since  $s_{2n+1} \rightarrow s$  and  $s_{2n} \rightarrow s$  as  $n \rightarrow +\infty$ , and consequently we have:

$$s_n \rightarrow s \quad \text{as } n \rightarrow +\infty$$



## Corollary 1: of the Leibniz test theorem

*Given a series satisfying the hypotheses of the Leibniz test theorem, we have:*

$$\underbrace{|s - s_m|}_{=|\sum_{k=m+1}^{\infty} (-1)^k a_k|} \leq a_{m+1}, \quad \forall m \in \mathbb{N}$$

- ▶ For every  $m$ , the error made by approximating  $s$  with  $s_m$  is, in absolute value, bounded above by the value of the first omitted term. In other words, the tail of the series tends to a value less than or equal to  $a_{m+1}$ .

# Proof

Since

$$s_{2n+1} \uparrow s \quad \text{and} \quad s_{2n} \downarrow s \quad \text{as } n \rightarrow +\infty$$

we therefore have, for every  $n \in \mathbb{N}$ :

$$s_{2n-1} \leq s \leq s_{2n} \quad \text{and} \quad s_{2n+1} \leq s \leq s_{2n}$$

from which we deduce

$$0 \leq s - s_{2n-1} \leq s_{2n} - s_{2n-1} = a_{2n}$$

$$0 \leq s_{2n} - s \leq s_{2n} - s_{2n+1} = a_{2n+1}$$

Therefore, for every  $m$ , whether even or odd, we have:

$$|s - s_m| = \left| \sum_{k=m+1}^{\infty} (-1)^k a_k \right| \leq a_{m+1}$$



## Example (1/2)

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} (-1)^k \frac{k-1}{k^2+k}$$

We have:

$$a_k = \frac{k-1}{k^2+k} = \frac{k-1}{k(k+1)} \geq 0, \forall k \in \mathbb{N}, k \geq 1$$

$$\frac{k-1}{k(k+1)} \sim \frac{1}{k} \rightarrow 0 \text{ as } k \rightarrow +\infty$$

hence the series does not converge absolutely.

## Example (2/2)

However, the series is decreasing for  $k \geq 2$  since:

$$\underbrace{\frac{k}{(k+1)(k+2)}}_{=a_{k+1}} \leq \underbrace{\frac{k-1}{k(k+1)}}_{=a_k} \iff k^2 \leq k^2 + k - 2 \iff k \geq 2$$

hence the series converges by the Leibniz test, since it is eventually decreasing.

## Example (1/2)

Let us determine the behavior of the series:

$$\sum_{k=2}^{\infty} (-1)^k \frac{\log k}{k}$$

We have:

$$a_k = \frac{\log k}{k} \geq 0, \forall k \geq 2 \quad \text{and} \quad \frac{\log k}{k} \rightarrow 0 \quad \text{as} \quad k \rightarrow +\infty$$

moreover

$$\frac{\log k}{k} > \frac{1}{k} \quad \text{for} \quad k \geq 3 \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k} \quad \text{diverges}$$

hence the series does not converge absolutely.

## Example (2/2)

Proving algebraically that the sequence  $\{a_k\}$  is decreasing is complicated; instead, we pass from the discrete to the continuous setting. We have, for  $x \in \mathbb{R}$  and  $x \geq 2$ :

$$f(x) = \frac{\log x}{x} \quad \text{and} \quad f'(x) = \frac{1 - \log x}{x^2} \leq 0 \quad \text{for} \quad x \geq e$$

It follows that  $f$  is decreasing for  $x \geq e$ ; consequently the sequence  $a_k = f(k)$  is decreasing for  $k \geq 3$  (the first integer  $> e$ ).

Hence the series converges by the Leibniz test.

## Sum of series

if  $\sum a_k$  converges and  $\sum b_k$  converges  
then  $\sum (a_k + b_k)$  converges

if  $\sum a_k$  converges and  $\sum b_k$  diverges  
then  $\sum (a_k + b_k)$  diverges

This can be checked by viewing the series as the limit of the sequence of partial sums and applying the theorem on the limit of a sum.

## Example (1/2)

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{k + 1 + (-1)^k k^2}{k^3}$$

We split it into the sum of two series:

$$\sum_{k=1}^{\infty} \frac{k + 1 + (-1)^k k^2}{k^3} = \sum_{k=1}^{\infty} \frac{k + 1}{k^3} + \sum_{k=1}^{\infty} \frac{(-1)^k}{k}$$

## Example (2/2)

For the first series we have:

$$\frac{k+1}{k^3} \geq 0, \forall k \geq 1 \quad \text{and} \quad \frac{k+1}{k^3} \rightarrow 0 \quad \text{as} \quad k \rightarrow +\infty$$

hence it converges by the limit comparison test, since:

$$\frac{k+1}{k^3} \sim \frac{1}{k^2} \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k^2} \quad \text{converges}$$

The second series converges by the Leibniz test; hence the original series converges.

## Example

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} (-1)^k \left( \frac{\sqrt{k} + (-1)^k}{k} \right)$$

We split it into the sum of two series:

$$\sum_{k=1}^{\infty} (-1)^k \left( \frac{\sqrt{k} + (-1)^k}{k} \right) = \sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}} + \sum_{k=1}^{\infty} \frac{1}{k}$$

The first series converges by the Leibniz test; the second diverges (harmonic series); hence the original series diverges.

## Example

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

We have:

$$\frac{(-1)^{k+1}}{k} = -\frac{(-1)^k}{k}, \quad \forall k \in \mathbb{N}, k > 1$$

moreover, the sequence  $a_k = \frac{1}{k}$  is decreasing and  $\frac{1}{k} \rightarrow 0$  as  $k \rightarrow +\infty$ , hence it satisfies the two conditions of the theorem.

Consequently, the series  $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$  converges.

# Summary

## Leibniz test:

- ▶  $\sum (-1)^k a_k$  with  $a_k \geq 0$  decreasing (also eventually) and  $a_k \rightarrow 0$  converges
- ▶  $s_{2n} \downarrow s$  (from above) and  $s_{2n+1} \uparrow s$  (from below)
- ▶ Error estimate:  $|s - s_m| \leq a_{m+1}$  (first omitted term)

## Techniques in the examples:

- ▶  $\sum \frac{(-1)^k}{k}$  converges but not absolutely
- ▶ Monotonicity via algebraic inequalities or passing to the continuous setting ( $f'(x) \leq 0$ )
- ▶ Splitting into a sum of series: convergent + convergent, convergent + divergent

# Final summary

- ▶ For series with terms of variable sign we first study **absolute convergence** with the tests for series with nonnegative terms
- ▶ Absolute convergence  $\Rightarrow$  simple convergence, but not conversely (counterexample:  $\sum \frac{(-1)^k}{k}$ )
- ▶ For alternating series we apply the **Leibniz test**:  $a_k \geq 0$ , decreasing and infinitesimal
- ▶ The corollary bounds the error made by truncating the series: at most the first omitted term
- ▶ A series can be split into a sum of simpler series studied separately